To evaluate the structural integrity of a launch vehicle nozzle coupled to a solid rocket motor casing under a maximum thrust load of 15 MN using ANSYS Mechanical. The analysis aims to determine the maximum von Mises stress, total deformation, and factor of safety under the applied loading and boundary conditions. Fixed supports are to be applied at the nozzle–casing coupling interfaces, and appropriate support configurations are to be investigated to achieve a target factor of safety of 5 while maintaining structural integrity.

DE-LAVEL Nozzle

Static Structural Analysis

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Structural Strength Analysis of Solid Rocket Motor Nozzle Under 15 MN Thrust

1. Objective

The objective of this analysis is to evaluate the structural strength of a solid rocket motor nozzle when subjected to a maximum thrust of **15 MN** and mechanically coupled to the solid motor casing of the launch vehicle.

The analysis is performed using ANSYS Static Structural to determine:

- Maximum Equivalent (von Mises) Stress
- Maximum Total Deformation
- Factor of Safety (FOS)

The required minimum Factor of Safety for the design is **5**.

The nozzle is manufactured from **Inconel 718**, while the solid motor casing is modelled using **Aluminium Alloy 7075-T6**.

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2. Geometry and Structural Configuration

The model consists of a convergent-divergent solid rocket motor nozzle coupled to the solid motor casing.

The nozzle incorporates a mechanical coupling arrangement consisting of **dovetail rings/slots**. The nozzle is inserted into the corresponding casing arrangement and rotated to engage the dovetail locking mechanism. Bolts are additionally provided to secure the connection.

The coupling arrangement transfers the structural reaction generated by the nozzle during operation into the solid motor casing and subsequently into the launch-vehicle structure.

The structural load path can therefore be represented as:

Nozzle → Dovetail Coupling → Bolted Joint → Aluminium 7075-T6 Motor Casing

For the present structural analysis, the detailed nonlinear behaviour of the dovetail and bolt connection is simplified through an appropriate boundary condition at the coupling faces.

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3. Material Properties

3.1 Nozzle — Inconel 718

The nozzle is assigned Inconel 718 material.

The principal material properties used for the structural analysis are:

Property	**Value**
Material	Inconel 718
Young's Modulus	~200 GPa
Poisson's Ratio	~0.29–0.30
Density	~8190 kg/m³
Yield Strength	~1035 MPa

The exact values used in the ANSYS Engineering Data are to be considered for the final numerical evaluation.

3.2 Motor Casing — Aluminium Alloy 7075-T6

The motor casing is assigned Aluminium Alloy 7075-T6.

Property	Value
Material	Aluminium Alloy 7075-T6
Young's Modulus	~71.7 GPa
Poisson's Ratio	~0.33
Density	~2810 kg/m³
Yield Strength	~503 MPa

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4. Loading Condition

The maximum operating thrust considered in the analysis is:

$$F = 15\ MN$$

Therefore,

$$F = 15{,}000{,}000\ N$$

The thrust-producing pressure acting on the internal surfaces of the nozzle is represented in the structural model by the corresponding pressure/load condition.

The applied loading represents the maximum thrust condition specified for the launch-vehicle solid motor.

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5. Boundary Conditions

The nozzle is mechanically coupled to the Aluminium 7075-T6 motor casing through the dovetail locking arrangement and bolts.

For the present analysis, **Fixed Support is applied at the faces of the nozzle-casing coupling** as specified in the problem statement.

The coupling faces represent the structural restraint provided by the locked nozzle-to-casing connection.

The Fixed Support prevents rigid-body translation and rotation of the nozzle at the coupling interface while allowing the structural response of the nozzle to be evaluated under the applied thrust loading.

No additional arbitrary supports are introduced solely to increase the calculated Factor of Safety. The support locations are selected based on the actual mechanical coupling arrangement.

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6. Meshing

A finite-element mesh is generated for the nozzle and motor-casing geometry.

Particular attention is given to the following regions:

- Nozzle throat
- Convergent section
- Divergent section

- Coupling region
- Dovetail transition regions
- Geometric discontinuities and sharp transitions

The mesh is refined in regions where significant stress gradients are expected.

A mesh-convergence check should be performed by progressively refining the mesh until the maximum stress and deformation results show acceptable variation.

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7. Analysis Procedure

The Static Structural analysis is performed using the following procedure:

1. Import/create the nozzle and motor-casing geometry.
2. Assign **Inconel 718** to the nozzle.
3. Assign **Aluminium Alloy 7075-T6** to the motor casing.
4. Define the mechanical coupling between the nozzle and casing.
5. Apply the maximum thrust loading corresponding to **15 MN**.
6. Apply Fixed Support to the appropriate coupling faces.
7. Generate the finite-element mesh.
8. Refine the mesh in the throat and coupling regions.
9. Solve the Static Structural analysis.
10. Obtain the Equivalent (von Mises) Stress distribution.
11. Obtain the Total Deformation distribution.
12. Determine the minimum Factor of Safety.
13. Compare the calculated Factor of Safety with the required value of **5**.

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8. Results and Discussion

8.1 Equivalent (von Mises) Stress

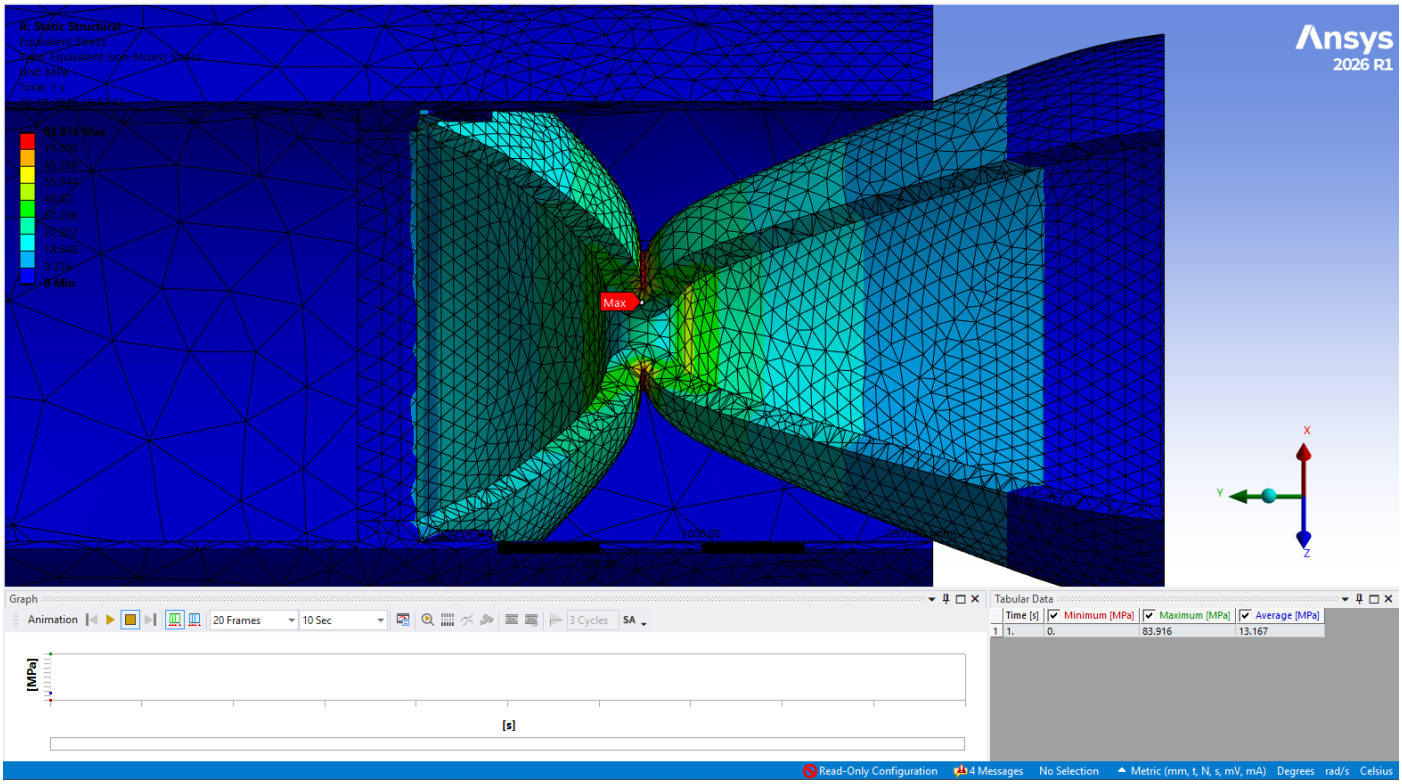
The maximum Equivalent (von Mises) Stress obtained from the structural analysis is:

$$\boxed{\sigma_{vm,max} = 83.916\ MPa}$$

The maximum stress occurs approximately at the **neck/throat region of the nozzle**.

The concentration of stress in this region is expected because the nozzle throat represents a significant geometric transition and experiences substantial structural loading.

The calculated stress is substantially below the assumed yield strength of Inconel 718.



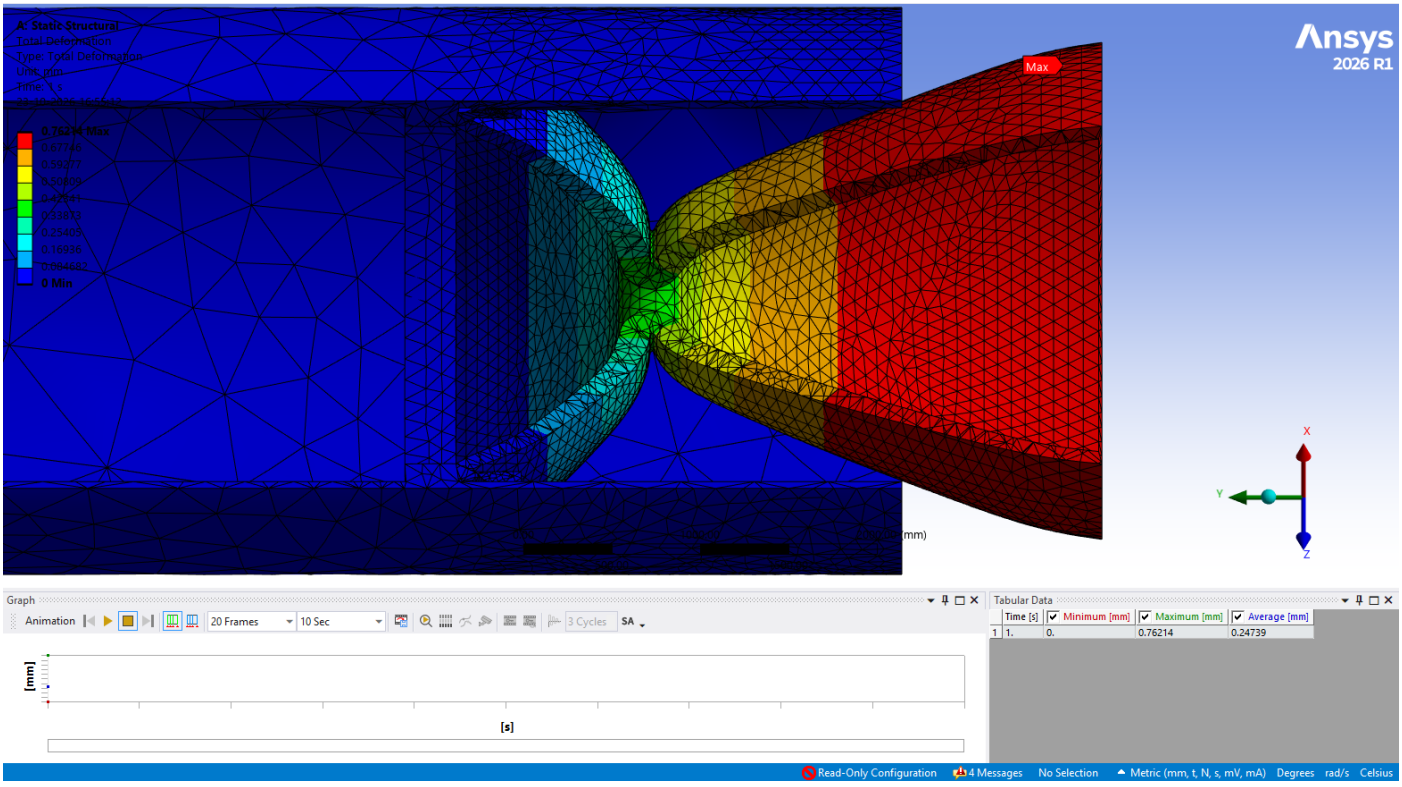
8.2 Total Deformation

The maximum Total Deformation obtained from ANSYS is:

$$\boxed{U_{max} = 0.76214\,mm}$$

The maximum deformation should be taken directly from the final ANSYS Total Deformation result.

The deformation result is used to verify that the nozzle remains within an acceptable structural displacement range under the maximum thrust condition.



8.3 Factor of Safety

The Factor of Safety is calculated using:

$$FoS = \frac{\sigma_y}{\sigma_{vm,max}}$$

For the assumed Inconel 718 yield strength:

$$\sigma_y = 1035\ MPa$$

and:

$$\sigma_{vm,max} = 83.916\ MPa$$

Therefore:

$$FoS = \frac{1035}{83.916}$$
$$\boxed{FoS \approx 12.337}$$

The required minimum Factor of Safety is:

$$FoS_{required} = 5$$

Therefore:

$$\boxed{12.337 > 5}$$

The nozzle therefore satisfies the specified minimum Factor of Safety requirement.

8.4 Allowable Stress Corresponding to FOS = 5

For a required Factor of Safety of 5, the corresponding allowable stress is:

$$\sigma_{allow} = \frac{\sigma_y}{FoS_{required}}$$
$$\sigma_{allow} = \frac{1035}{5}$$
$$\boxed{\sigma_{allow} \approx 207\ MPa}$$

The calculated maximum stress is:

$$83.916\ MPa$$

which is significantly lower than the allowable stress of approximately 207 MPa.

Thus:

$$83.916 < 207\ MPa$$

and the required safety margin is satisfied.

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9. Interpretation of the Support Requirement

The instruction to:

"Provide as many supports as needed till the safety factor margin reaches 5"

is interpreted as requiring sufficient structural restraint to represent the actual nozzle-to-casing connection and prevent rigid-body motion.

The supports are therefore applied at the physical coupling faces rather than at arbitrary locations on the nozzle.

The purpose of the support is to represent the reaction supplied by the dovetail and bolted connection between the nozzle and the solid motor casing.

Since the resulting Factor of Safety is approximately **12.3**, which is already greater than the required value of **5**, additional artificial supports are not required solely to increase the Factor of Safety.

Adding unnecessary Fixed Supports to other nozzle surfaces would artificially restrict the deformation and could produce an unrealistic stress distribution.

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10. Summary of Results

Parameter	**Result**
Maximum Thrust	**15 MN**
Nozzle Material	**Inconel 718**
Motor Casing Material	**Aluminium 7075-T6**
Maximum von Mises Stress	**83.916 MPa**
Maximum Total Deformation	**0.76214 mm**
Yield Strength Used for Nozzle	**1035 MPa**
Calculated Factor of Safety	**12.337**
Required Factor of Safety	**5**
Requirement	**Satisfied**

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11. Conclusion

A Static Structural analysis of the solid rocket motor nozzle was performed for a maximum thrust condition of **15 MN**.

The nozzle was modelled using **Inconel 718**, while the associated motor casing was modelled using **Aluminium Alloy 7075-T6**. The nozzle is mechanically coupled to the casing using a dovetail locking arrangement and bolts. For the structural analysis, the coupling faces were restrained using Fixed Support to represent the mechanical restraint provided by the locked connection.

The maximum Equivalent (von Mises) Stress obtained from the analysis was:

$$\boxed{83.916\ MPa}$$

and was located near the nozzle throat/neck region.

Using an assumed Inconel 718 yield strength of approximately **1035 MPa**, the calculated Factor of Safety was:

$$\boxed{FoS = 12.337}$$

which is greater than the required minimum value of:

$$\boxed{FoS = 5}$$

Therefore, the nozzle satisfies the specified structural safety requirement under the 15 MN maximum thrust loading condition.

The maximum deformation obtained from ANSYS should be included in the final results table along with its corresponding location.

Overall, the analysis indicates that the nozzle has sufficient structural strength for the specified loading condition under the stated material properties and boundary-condition assumptions.